Neural Multichannel Distant Speaker Diarization and Source Separation with Beta Speaker Activity Prior

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Background

Speaker diarization

Figure out *who spoke when* in a multi-speaker audio recording.

Speech signals in meeting scenario

- Data: far field, multi-channel speech mixture signals
- Challenges: noise, reverberation; overlap speeches; unknown number of speakers

Methods

- Data-driven:
  - Large data for end-to-end training (e.g. EEND [Fujita et al., 2020])
  - Large parameter with LLM backbone (e.g. VibeVoice-ASR [Peng et al., 2026])
- Model-driven: Incorporate prior knowledge to reduce the data and parameter requirements.

Diarization meets separation

A close relationship between source separation and speaker diarization:

- If separation is perfect, diarization reduces to VAD.
- If diarization is perfect, separation benefits from the speaker activity mask.

A baseline model: Neural FCASA[Bando et al., 2024]

- Stands for Neural **F**ull-rank **C**ovariance **A**nalysis with **S**peaker **A**ctivity.
- Marrying a source separation model and a diarization model to do joint task
- Compared to the data-driven models, it is data-efficient (~100h) and parameter-efficient(~20M)

From a separation perspective, it is a source separation model with diarization annotations as weak supervision. [Bando et al., 2024]

From a diarization perspective, it is a diarization model incorporating prior knowledge from a source separation model. [Our work]

Generative modeling in Neural FCASA

A deep latent source feature

$$\mathbf{z}_{nt} \sim \mathcal{N}\left(\mathbf{0}, \mathbf{I}\right),$$

that generates the source power spectrum density (PSD)

$$\lambda_{nft} = g_{\theta,f}(\mathbf{z}_{nt}),$$

that generates the mixture signal PSD, with spatial covariance matrix and the speaker activity mask

$$\mathbf{Y}_{ft} = \sum_{n=1}^{N} u_{nt} \underbrace{\lambda_{nft}\mathbf{R}_{nf}}_{\triangleq \mathbf{Y}_{nft}},$$

Finally the mixture signal is realized as Gaussian variance modeling of the PSD

$$\mathbf{x}_{ft} \sim \mathcal{N}_{\mathbb{C}}\left(\mathbf{0}, \mathbf{Y}_{ft}\right).$$

Objective

The objective of joint diarization and separation task is formalized as solving two probabilitic models:

$$\text{Diarization: } p(\mathbf{U} \mid \mathbf{X}) \text{ and Separation: } p(\mathbf{Z} \mid \mathbf{X})$$

However it is intractable to learn $p(\mathbf{x})$ -> variational inference.

Proposing two surrogate models

$$q_\phi(\mathbf{Z} \mid \mathbf{X}) \triangleq \prod_{n,t,d=1}^{N,T,D} \mathcal{N}(z_{ntd} \; ; \; \mu_{\phi,ntd}, \sigma^2_{\phi,ntd}),$$
$$q_\phi(\mathbf{U} \mid \mathbf{X}) \triangleq \prod_{n,t=1}^{N,T} \text{Bernoulli}(u_{nt}; \eta_{\phi,nt}),$$

and computing a tractable evidence lower bound (ELBO) for the likelihood.

Variational Inference

$$\log p_\theta(\mathbf{X}, \mathbf{U}) \geq \underbrace{\mathbb{E}_{q_\phi(\mathbf{Z}|\mathbf{X})}[\log p_\theta(\mathbf{X} \mid \mathbf{U}, \mathbf{Z})]}_{\mathcal{L}_{recons}^{(sep)}} \underbrace{-\mathrm{KL}[q_\phi(\mathbf{Z} \mid \mathbf{X}) \mid\mid p_\theta(\mathbf{Z})]}_{\mathcal{L}_{kl}^{(sep)}} + \underbrace{\log p(\mathbf{U})}_{\mathcal{L}^{(diar)}},$$

$\mathcal{L}_{recons}^{(sep)}$ and $\mathcal{L}_{kl}^{(sep)}$: ELBO [Kingma et al., 2014a] for separation model, with a teacher-forcing trick ($\mathbf{U}$).

$\mathcal{L}^{(diar)}$: $\log p(\mathbf{U})$ is not defined in the generative model, instead, use $\log q_\phi(\mathbf{U}|\mathbf{X})$ as an approximation. Similar to semi-supervised VAE loss [Kingma et al., 2014b].

Can we define a prior model for the speaker label $\mathbf{U}$, so that we can perform variational inference for the diarization model as well?

Prior model for speaker activity

A prior model for the speaker activity mask $\mathbf{U}$, which might intuitively capture prior knowledge about the speaker activity patterns, including:

1. the speaker' s character, e.g., whether he/she is outgoing or shy to speak,
2. the ambiance of the conversation, e.g., whether it is encouraging or discouraging for the attendees to speak.

As the speaker activity mask $\mathbf{U}$ is a binary variable, we propose to model it as a Bernoulli variable with a Beta prior:

$$\eta_{nt} \sim \mathrm{Beta}(\alpha, \beta)$$
$$u_{nt} \sim \mathrm{Bernoulli}(\eta_{nt})$$

When combined with neural FCASA, we call it **Beta FCASA**.

Variantional Inference in Beta FCASA

The inference model:

$$q_\phi(\mathbf{H} \mid \mathbf{X}) \triangleq \prod_{n,t=1}^{N,T} \text{Beta}\left(\eta_{nt} \mid \alpha_{\phi,nt}, \beta_{\phi,nt}\right),$$

The ELBO for the joint likelihood now becomes:

$$\log p_\theta(\mathbf{X}, \mathbf{U}) \geq \underbrace{\mathbb{E}_{q_\phi(\mathbf{Z}|\mathbf{X},\mathbf{U})}[\log p_\theta(\mathbf{X} \mid \mathbf{U}, \mathbf{Z})]}_{\mathcal{L}_{recons}^{(sep)}} \underbrace{-D_{\mathrm{KL}}[q_\phi(\mathbf{Z} \mid \mathbf{X}) \| p_\theta(\mathbf{Z})]}_{\mathcal{L}_{kl}^{(sep)}}$$
$$+ \underbrace{\mathbb{E}_{q_\phi(\mathbf{H})}[\log p_\theta(\mathbf{U} \mid \mathbf{H})]}_{\mathcal{L}_{recons}^{(diar)}} \underbrace{-D_{\mathrm{KL}}(q_\phi(\mathbf{H}) \| p_\theta(\mathbf{H}))}_{\mathcal{L}_{kl}^{(diar)}},$$

i.e., for the diarization likelihood, we optimize explicitly its ELBO function.

Tractablitity of the ELBO

- The separation part: standard (Gaussian) VAE ELBO, tractable (See VAE paper [Kingma et al., 2014a])
- The diarization part:
  - The reconstruction loss: use Beta-Bernoulli conjugacy, tractable

$$u\psi(\alpha_\phi) + (1-u)\psi(\beta_\phi) - \psi(\alpha_\phi + \beta_\phi)$$

  - The KL divergence: use the closed-form of KL divergence between two Beta distributions, tractable

$$\ln \frac{\Gamma(\alpha_\phi + \beta_\phi)\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)\Gamma(\alpha_\phi)\Gamma(\beta_\phi)} + (\alpha_\phi - \alpha)\psi(\alpha_\phi) + (\beta_\phi - \beta)\psi(\beta_\phi) - (\alpha_\phi + \beta_\phi - \alpha - \beta)\psi(\alpha_\phi + \beta_\phi)$$

  where $\psi(\cdot)$ is the digamma function and $\Gamma(\cdot)$ is the gamma function.

A subtlety of the Beta distribution

Under the canonical parametrization, the PDF of Beta distribution is:

$$\mathrm{Beta}(u;\alpha,\beta) = \frac{u^{\alpha-1}(1-u)^{\beta-1}}{B(\alpha,\beta)}, \quad u \in (0,1)$$

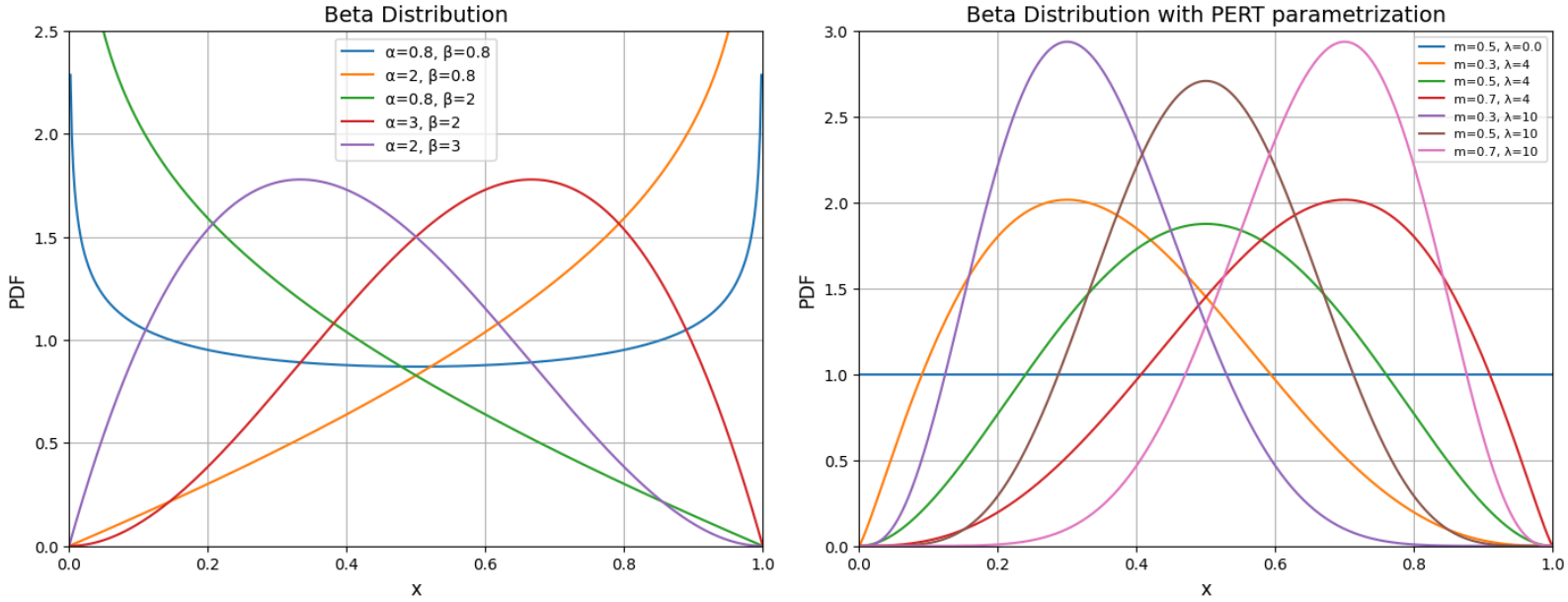
where the shape parameters α and β has no interpretable statistical meaning. Instead, we use the PERT parametrization (Malcolm et al. 1959), which is a reparametrization of the Beta distribution in terms of its *mode* m and *concentration* λ:

$$\begin{aligned}\mathrm{PERT} &: (\alpha,\beta) \mapsto \left(\frac{\alpha-1}{\alpha+\beta-2}, \alpha+\beta-2\right), \\ \mathrm{PERT}^{-1} &: (m,\lambda) \mapsto (1+\lambda m, 1+\lambda(1-m))\end{aligned}$$

This is a differentiable bijection for $\alpha, \beta > 1$.

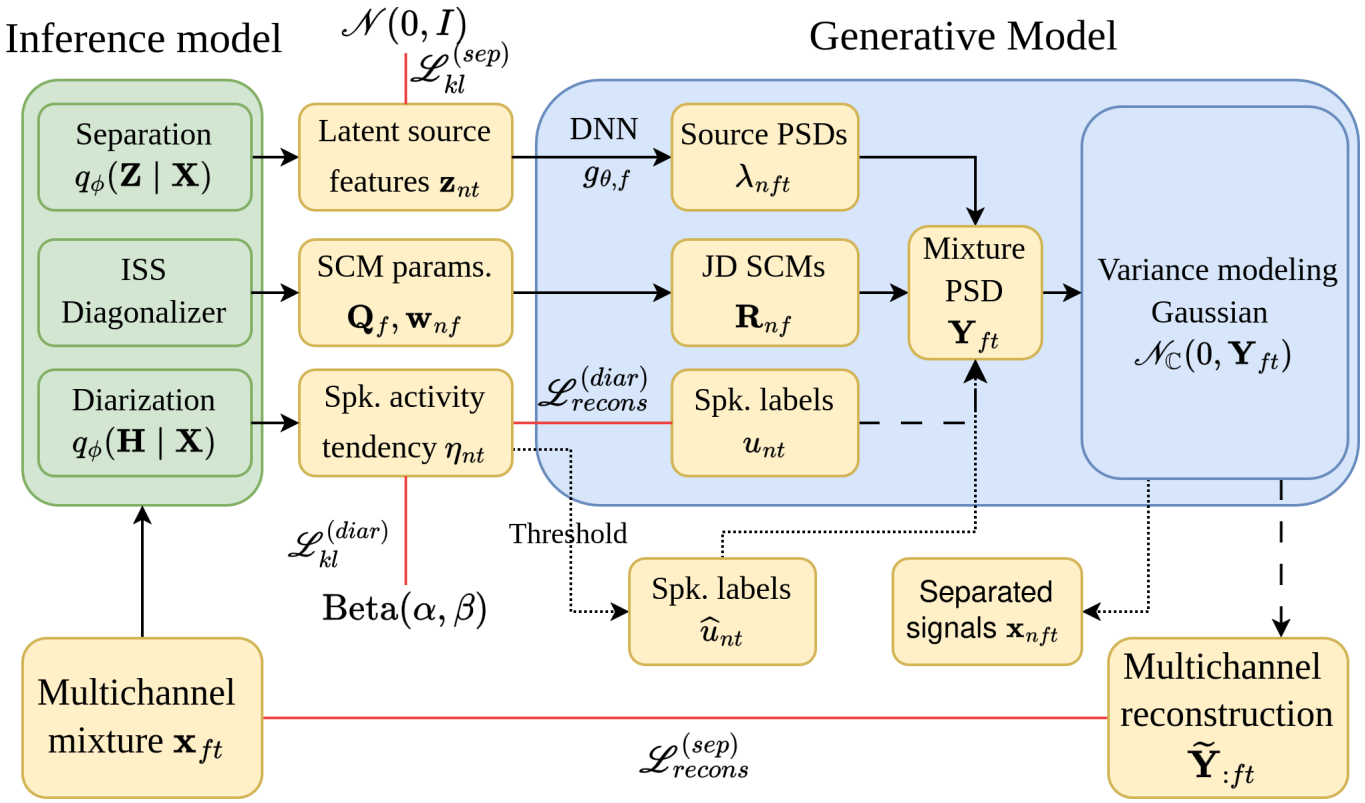
A subtlety of the Beta distribution

An illustration of the shape of Beta distribution w.r.t. its parametrization:



In short, under PERT parametrization, we can directly optimize the shape.

Wrap-up


Inference model
$\mathcal{N}(0, I)$
$\mathcal{L}_{kl}^{(sep)}$
Generative Model
Separation $q_\phi(\mathbf{Z} \mid \mathbf{X})$
ISS Diagonalizer
Diarization $q_\phi(\mathbf{H} \mid \mathbf{X})$
Latent source features $\mathbf{z}_{nt}$
SCM params. $\mathbf{Q}_f, \mathbf{w}_{nf}$
Spk. activity tendency η_{nt}
DNN $g_{\theta,f}$
Source PSDs λ_{nft}
JD SCMs $\mathbf{R}_{nf}$
Spk. labels u_{nt}
Mixture PSD $\mathbf{Y}_{ft}$
Variance modeling Gaussian $\mathcal{N}_{\mathbb{C}}(0, \mathbf{Y}_{ft})$
$\mathcal{L}_{recons}^{(diar)}$
$\mathcal{L}_{kl}^{(diar)}$
Beta(α, β)
Threshold
Spk. labels $\hat{u}_{nt}$
Separated signals $\mathbf{x}_{nft}$
Multichannel mixture $\mathbf{x}_{ft}$
Multichannel reconstruction $\tilde{\mathbf{Y}}_{:ft}$
$\mathcal{L}_{recons}^{(sep)}$

Results

m, λ	Setup	Miss (↓)	FA (↓)	Conf. (↓)	DER (↓)	JER (↓)
(baseline)	Forgiving	5.59	8.16	0.73	14.48	13.50
	Fair	8.61	6.33	0.79	15.73	23.16
	Full	10.77	6.81	1.16	18.73	27.65
	Overlap	19.40	3.80	0.90	24.11	26.53
0.3, 4	Forgiving	5.37	**4.31**	**0.43**	**10.11**	**9.73**
	Fair	8.00	**3.39**	**0.44**	**11.84**	**16.85**
	Full	9.96	**4.70**	**0.65**	15.31	**21.98**
	Overlap	17.20	**2.71**	**0.49**	20.41	**21.90**
0.5, 4	Forgiving	**4.55**	5.89	0.50	10.94	9.93
	Fair	**6.97**	4.69	0.53	12.19	17.04
	Full	**8.70**	6.07	0.79	15.55	22.21
	Overlap	**15.23**	3.47	0.64	**19.35**	20.96
0.7, 4	Forgiving	5.16	4.64	0.49	10.29	9.97
	Fair	7.69	3.70	0.50	11.89	17.40
	Full	9.69	4.77	0.75	**15.21**	22.57
	Overlap	17.11	2.80	0.56	20.48	22.33
Pyannote 3.1 IDM	Full	9.5	3.6	5.7	18.8	-
Pyannote 3.1 SDM	Full	11.2	3.8	7.5	22.4	-

Table 1: *Diarization metrics of the trained models on AMI evaluation set. All metrics are reported in percentage with the best score in **bold** and the second-best score underlined. We also list the single channel diarization baseline Pyannote 3.1 [29] as a reference.*

We choose different m, λ for the prior model $p(\mathbf{H})$, and verify our methods on the AMI dataset.

- We observe consistently better performance than the baseline model (without Beta prior).
- λ has less impact on the results, comparing to m.
- Following experiments on AliMeeting and CHiME-6 demonstrates the efficacy of our methods on broader datasets (Not shown in the paper but in my thesis)

Future work

- The estimation of the optimal hyperparameter m and λ.
- A better temporal model for speaker activity that incorporates more complex conversational dynamics.
- Improving diarization with a better separation model (Another work of us with heavy-tailed separation model, to be presented on SLT 2026 😊).

Thank you!